

and likely and the gains less likely but more extreme. Therefore, if skewness is positive, the average magnitude of positive deviations is larger than the average magnitude of negative deviations.

The approximation for computing **sample skewness** when n is large (100 or more) is:

$$\text{Skewness} \approx \left(\frac{1}{n}\right) \frac{\sum_{i=1}^n (X_i - \bar{X})^3}{s^3}. \quad (8)$$

As you will learn later in the curriculum, different investment strategies may introduce different types and amounts of skewness into returns.

Kurtosis

Another way in which a return distribution might differ from a normal distribution is its relative tendency to generate large deviations from the mean. Most investors would perceive a greater chance of extremely large deviations from the mean as higher risk.

Kurtosis is a measure of the combined weight of the tails of a distribution relative to the rest of the distribution—that is, the proportion of the total probability that is outside of, say, 2.5 standard deviations of the mean. A distribution that has fatter tails than the normal distribution is referred to as **leptokurtic** or **fat-tailed**; a distribution that has thinner tails than the normal distribution is referred to as being **platykurtic** or **thin-tailed**; and a distribution similar to the normal distribution as it concerns relative weight in the tails is called **mesokurtic**. A fat-tailed (thin-tailed) distribution tends to generate more frequent (less frequent) extremely large deviations from the mean than the normal distribution.

Exhibit 18 illustrates a distribution with fatter tails than the normal distribution. By construction, the fat-tailed and normal distributions in Exhibit 18 have the same mean, standard deviation, and skewness. Note that this fat-tailed distribution is more likely than the normal distribution to generate observations in the tail regions defined by the intersection of the distribution lines near a standard deviation of about ± 2.5 . This fat-tailed distribution is also more likely to generate observations that are near the mean, defined here as the region ± 1 standard deviation around the mean. However, to ensure probabilities sum to 1, this distribution generates fewer observations in the regions between the central region and the two tail regions.

Exhibit 18: Fat-Tailed Distribution Compared to the Normal Distribution

